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In [5]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset

df = pd.read_csv('laptop_price.csv', encoding='latin-1')

# Derived numeric columns for easier analysis
# (these are created from existing text columns in the dataset)
df['Weight_kg'] = df['Weight'].astype(str).str.replace('kg', '', regex=False).astype(float)
df['Ram_GB'] = df['Ram'].astype(str).str.replace('GB', '', regex=False).astype(int)

df.head()
```

Out[5]:

	laptop_ID	Company	Product	TypeName	Inches	ScreenResolution	Cpu	Ram	Memory	Gpu	OpSys	Weight	Price_euros
0	1	Apple	MacBook Pro	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 2.3GHz	8GB	128GB SSD	Intel Iris Plus Graphics 640	macOS	1.37kg	1339.69
1	2	Apple	Macbook Air	Ultrabook	13.3	1440x900	Intel Core i5 1.8GHz	8GB	128GB Flash Storage	Intel HD Graphics 6000	macOS	1.34kg	898.94
2	3	HP	250 G6	Notebook	15.6	Full HD 1920x1080	Intel Core i5 7200U 2.5GHz	8GB	256GB SSD	Intel HD Graphics 620	No OS	1.86kg	575.00
3	4	Apple	MacBook Pro	Ultrabook	15.4	IPS Panel Retina Display 2880x1800	Intel Core i7 2.7GHz	16GB	512GB SSD	AMD Radeon Pro 455	macOS	1.83kg	2537.45
4	5	Apple	MacBook Pro	Ultrabook	13.3	IPS Panel Retina Display 2560x1600	Intel Core i5 3.1GHz	8GB	256GB SSD	Intel Iris Plus Graphics 650	macOS	1.37kg	1803.60

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In [10]: # 1) Central tendency of data
price = df['Price_euros']
mean_price = price.mean()
median_price = price.median()
mode_price = price.mode().iloc[0]

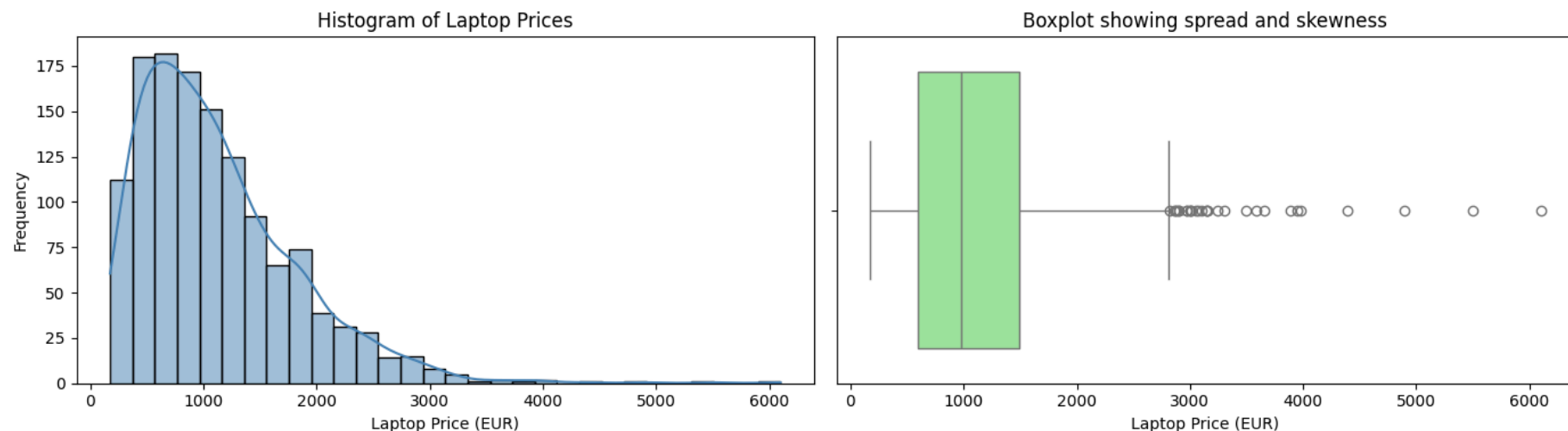
plt.figure(figsize=(14, 4))
ax1 = plt.subplot(1, 2, 1)
sns.histplot(price, bins=30, kde=True, color='steelblue', ax=ax1)
ax1.set_title('Histogram of Laptop Prices')
ax1.set_xlabel('Laptop Price (EUR)')
ax1.set_ylabel('Frequency')

ax2 = plt.subplot(1, 2, 2)

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sns.boxplot(x=price, color='lightgreen', ax=ax2)
ax2.set_title('Boxplot showing spread and skewness')
ax2.set_xlabel('Laptop Price (EUR)')
plt.tight_layout()
plt.show()

print('Central tendency summary for Price_euros:')
print(f'Mean: {mean_price:.2f}')
print(f'Median: {median_price:.2f}')
print(f'Mode: {mode_price:.2f}')
```



Central tendency summary for Price_euros:

Mean: 1123.69

Median: 977.00

Mode: 1099.00

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In [6]: # 2) Outlier identification (visual + numeric)
Q1 = price.quantile(0.25)
Q3 = price.quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
outliers = df[(price < lower_bound) | (price > upper_bound)].copy()

print('\nOutlier detection summary:')
print(f'Q1: {Q1:.2f}')
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print(f'Q3: {Q3:.2f}')
print(f'IQR: {IQR:.2f}')
print(f'Lower bound: {lower_bound:.2f}')
print(f'Upper bound: {upper_bound:.2f}')
print(f'Number of outliers: {len(outliers)}')

plt.figure(figsize=(8, 4))
sns.boxplot(y='Price_euros', data=df, color='lightcoral')
plt.title('Boxplot for identifying outliers in Laptop Price')
plt.ylabel('Price (EUR)')
plt.show()

print('\nOutlier sample rows:')
print(outliers[['Company', 'Product', 'TypeName', 'Price_euros', 'Ram_GB', 'Weight_kg']].head(10).to_string(index=False))
```

Outlier detection summary:

Q1: 599.00

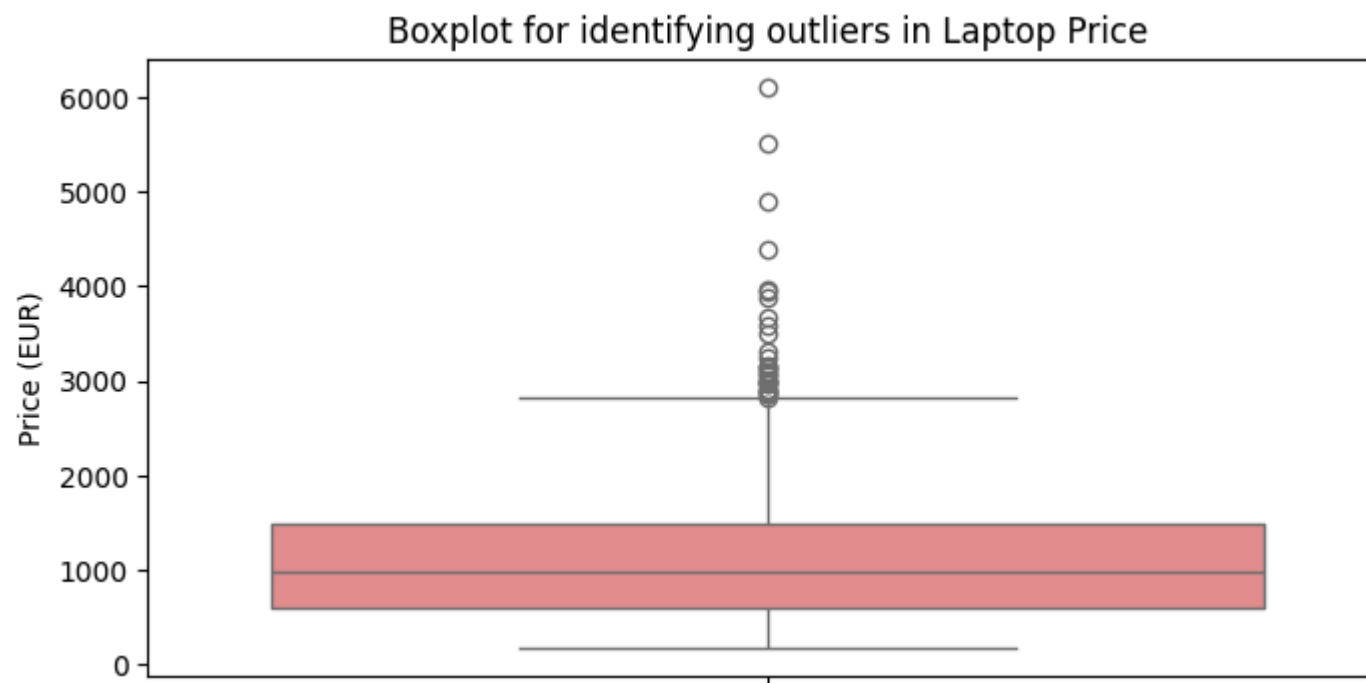
Q3: 1487.88

IQR: 888.88

Lower bound: -734.32

Upper bound: 2821.20

Number of outliers: 29

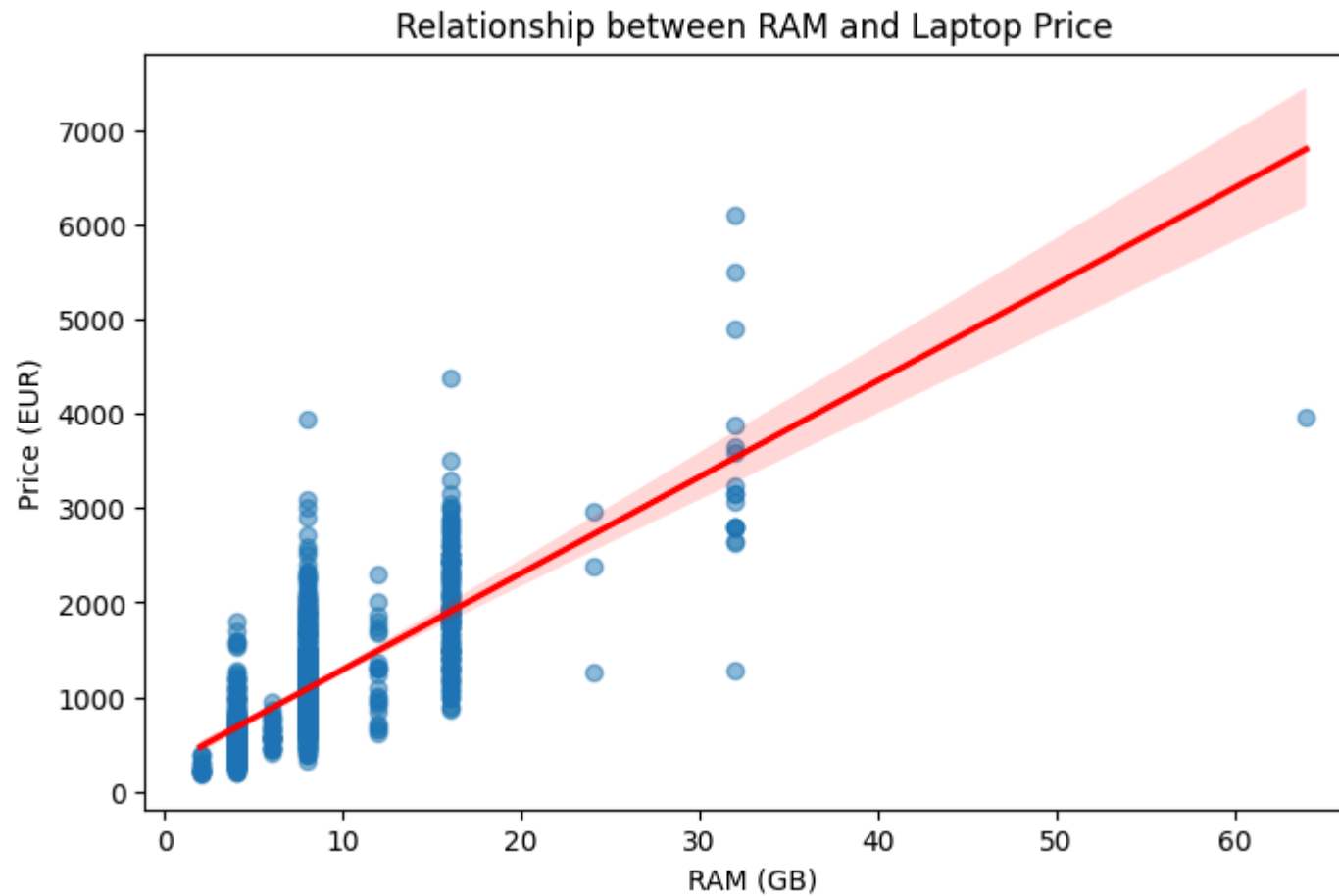


Outlier sample rows:

Company	Product	TypeName	Price_euros	Ram_GB	Weight_kg
Apple	MacBook Pro	Ultrabook	2858.00	16	1.83
Lenovo	Thinkpad Yoga 2 in 1	Convertible	2824.00	16	1.42
Razer	Blade Pro	Gaming	6099.00	32	3.49
Dell	Precision 7520	Workstation	3055.00	16	2.80
Asus	ROG G703VI-E5062T	Gaming	3890.00	32	4.70
Asus	Rog G701VIK-BA060T	Gaming	2999.00	16	3.60
Dell	Precision 7720	Workstation	2884.86	16	3.42
Asus	ROG Zephyrus	Gaming	2968.00	24	2.24
Dell	Alienware 17	Gaming	3012.77	16	4.42
Lenovo	Thinkpad P71	Notebook	2999.00	8	3.40

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In [7]: # 3) Relationship between two numeric variables
plt.figure(figsize=(8, 5))
sns.regplot(data=df, x='Ram_GB', y='Price_euros', scatter_kws={'alpha': 0.5}, line_kws={'color': 'red'})
plt.title('Relationship between RAM and Laptop Price')
plt.xlabel('RAM (GB)')
plt.ylabel('Price (EUR)')
plt.show()
```

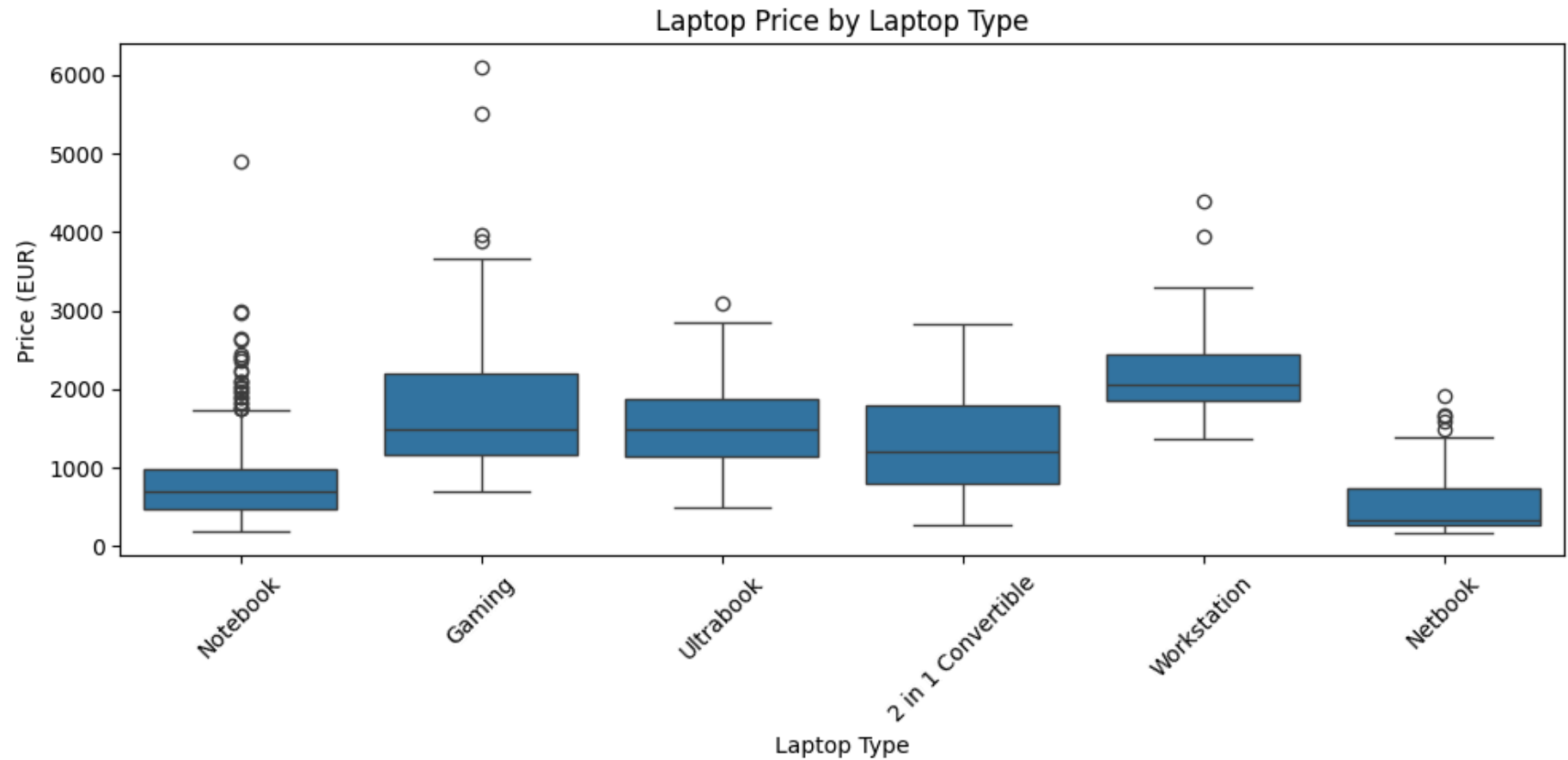
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corr = df[['Ram_GB', 'Price_euros']].corr().iloc[0, 1]
print(f'\nCorrelation between RAM and Price: {corr:.2f}')
```



Correlation between RAM and Price: 0.74

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In [8]: # 4) Relationship between numeric and nominal variable
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='TypeName', y='Price_euros', order=df['TypeName'].value_counts().index)
plt.title('Laptop Price by Laptop Type')
plt.xlabel('Laptop Type')
plt.ylabel('Price (EUR)')
plt.xticks(rotation=45)
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plt.tight_layout()
plt.show()
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In [9]: # 5) Relationship between two nominal variables
company_type = pd.crosstab(df['Company'], df['TypeName'])
plt.figure(figsize=(12, 8))
sns.heatmap(company_type, cmap='YlGnBu', annot=False, fmt='d')
plt.title('Company vs Laptop Type (Count Heatmap)')
plt.xlabel('Laptop Type')
plt.ylabel('Company')
plt.tight_layout()
plt.show()
```

